

7.1.2.1.3 Interoperability across and within FRMCS Domains is one of the underlying design principles of this document. (I)

7.1.2.2 Design Principles

7.1.2.2.1 The separation of Strata (as described in the present document) is a design principle of the On-Board FRMCS. (I)

Note: The TOBA architecture refers only to Service and Transport Strata, not to the Application Stratum.

7.1.2.2.2 Modularity is a design principle of the On-Board FRMCS. (I)

7.1.2.2.3 Scalability is a design principle of the On-Board FRMCS. (I)

7.1.2.2.4 Vendor diversity of Radio Modules is a design principle of the On-Board FRMCS. (I)

7.1.2.2.5 Interchangeability is a design principle of the On-Board FRMCS. (I)

7.1.2.2.6 Distributed Architecture is a design principle of the On-Board FRMCS. (I)

7.1.2.2.7 Untrusted environment for local communication is a design characteristic of the On-Board FRMCS. (I)

7.1.3 On-Board FRMCS Architecture overview

7.1.3.1 Introduction

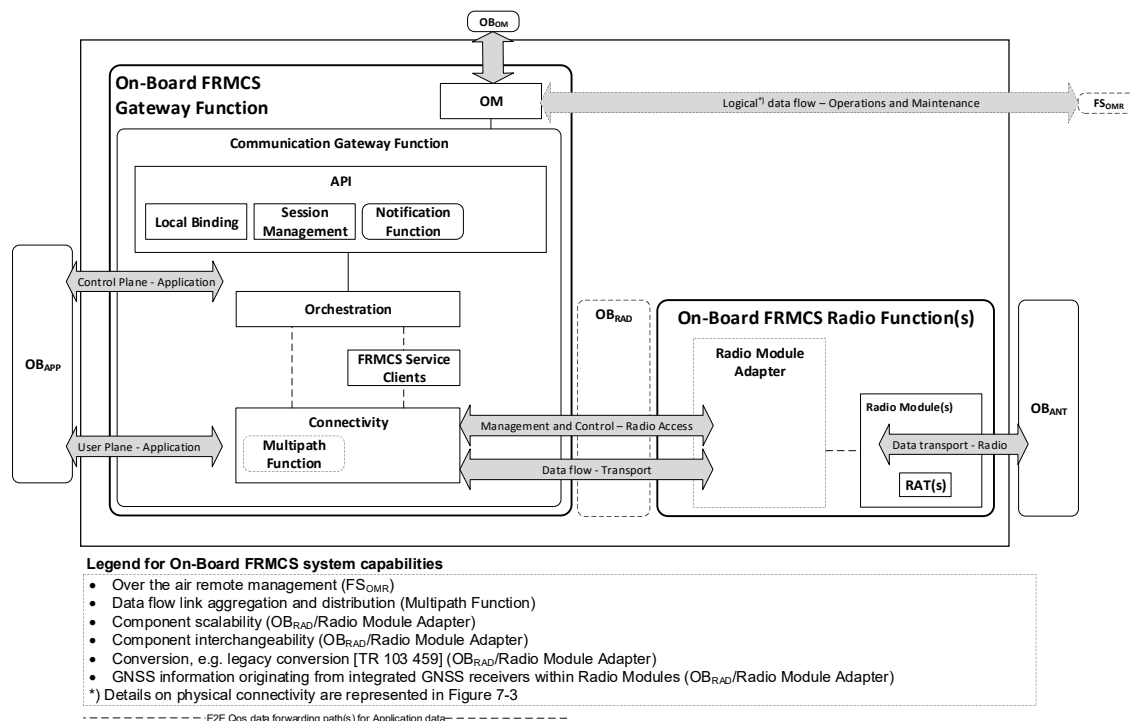


Figure 7-2: On-Board FRMCS architecture

7.1.3.1.1 The figure 7-2 represents the On-Board FRMCS Architecture which, together with its associated system requirements, describes the On-Board FRMCS v2. (I)

7.1.3.1.2 The schematic in Figure 7-2: is intended to support (at least) the following (I):

- Integrated architecture
- Integrated architecture providing interchangeability
- Distributed architecture providing interchangeability

7.1.3.2 Architecture scope

7.1.3.2.1 The On-Board FRMCS includes the following functions (I):

- On-Board FRMCS Gateway Function ([7.1.4.2], [7.1.5])
- On-Board FRMCS Radio Function ([7.1.4.2], [7.1.5.10.2.4], [7.1.6])

7.1.3.2.2 The On-Board FRMCS Gateway Function includes the following functions (I):

- Communication Gateway Function ([7.1.5.3])
- OM ([7.1.5.11])

7.1.4 Interfaces

7.1.4.1 External Interfaces

7.1.4.1.1 OB_{APP}

7.1.4.1.1.1 The OB_{APP} interface is developed more in detail in section [6.4.1.1] of this document and in [FRMCS-FFFIS]. (I)

7.1.4.1.1.2 Physical Interface between On-Board FRMCS and the applications (OB_{APP})

7.1.4.1.1.2.1 Rationale

The on-board applications need to have connectivity in order to utilize the On-Board FRMCS. This connectivity can be of different type(s) depending on the user equipment where on-board application(s) are installed. e.g. handheld device(s) for user(s) moving within or in close proximity to a train, fixed equipment within a train or mounted on the train. (I)

7.1.4.1.1.2.2 Requirements

7.1.4.1.1.2.2.1 The On-Board FRMCS shall provide wired connectivity for devices using off the shelf technologies (e.g., Ethernet). (M)

7.1.4.1.1.2.2.2 Intentionally deleted.

7.1.4.1.1.2.2.3 The number of physical interfaces for OB_{APP} connectivity should be expandable. (O)

7.1.4.1.2 OB_{ANT}

7.1.4.1.2.1 Description

7.1.4.1.2.1.1 Interface OB_{ANT} is an implementation of the OB_{ANT} reference point. This interface enables communication over the air gap between On-Board FRMCS and Tracksides. (I)

7.1.4.1.2.1.2 OB_{ANT} implements connection of GNSS receiver and GNSS antenna. (I)

7.1.4.1.2.2 Interface Requirements

7.1.4.1.2.2.1 For each On-Board FRMCS Radio Module, the On-Board FRMCS shall provide OB_{ANT} interfaces at the OB_{ANT} reference point. (M)

7.1.4.1.2.2.2 The On-Board FRMCS shall enable wireless Communication Services through interfaces exposed at the OB_{ANT} reference point. (M)

7.1.4.1.2.2.3 OB_{ANT} should be implemented using suitable, current and widely adopted industry specifications. (I)

7.1.4.1.2.3 Physical Interface between FRMCS Radio Modules and the antenna system (OB_{ANT})

7.1.4.1.2.3.1 Rationale

7.1.4.1.2.3.1.1 An industry-specified interface between the FRMCS Radio Modules and the antenna system, facilitates a modular implementation of the FRMCS On-Board System and increases the degrees of freedom for the fleet managers during the life cycle of the FRMCS On-Board System. (I)

7.1.4.1.2.3.2 Requirements

7.1.4.1.2.3.2.1 The OB_{ANT} physical interface shall be realised using a suitable existing industry specification. (M)

7.1.4.1.3 OB_{OM}

7.1.4.1.3.1 Description

7.1.4.1.3.1.1 Interface OB_{OM} is an implementation of the OB_{OM} reference point. (I)

7.1.4.1.3.1.2 Interface OB_{OM} enables on-board communication between Operation and Maintenance System and On-Board FRMCS. (I)

7.1.4.1.3.2 Interface Requirements

7.1.4.1.3.2.1 The On-Board FRMCS shall provide external interfaces at the OB_{OM} reference point. (M)

Note: OM and associated interfaces (OB_{OM} and FS_{OMR}) are left for implementation in V3.

7.1.4.1.3.2.2 The On-Board FRMCS shall enable Operations and Maintenance by enabling the OB_{OM} interface. (M)

7.1.4.1.3.3 Physical Interface between local management system(s) and O&M Function of the On-Board FRMCS (OB_{OM})

7.1.4.1.3.3.1 Rationale

7.1.4.1.3.3.1.1 An industry specified physical interface between the O&M Function of the On-Board FRMCS and local management system(s) will enable and promote flexible implementation and business models in terms of fleet management with respect to the management of On-Board FRMCS. (I)

7.1.4.1.3.3.2 Requirements

7.1.4.1.3.3.2.1 The On-Board FRMCS shall provide connectivity for local O&M access using off the shelf technologies e.g. Ethernet. (M)

7.1.4.1.4 FS_{OMR}

7.1.4.1.4.1 Description

7.1.4.1.4.1.1 Interface FS_{OMR} is implemented over the FS_{OMR} reference point as defined in section [6.4.3.1] of this document. (I)

Note: The FS_{OMR} reference point is not a reference point belonging to the On-Board FRMCS scope, nevertheless the use of FS_{OMR} interface has dependencies to the On-Board FRMCS operations and maintenance.

7.1.4.1.4.1.2 Interface FS_{OMR} enables over the air gap communication between Operation and Maintenance System and On-Board FRMCS. (I)

7.1.4.1.4.2 Interface Requirements

7.1.4.1.4.2.1 The FRMCS System shall expose external interfaces at the FS_{OMR} reference point. (M-V3)

7.1.4.1.4.2.2 The On-Board FRMCS shall enable Operations and Maintenance using the FS_{OMR} interface enabled by the FRMCS System by using over the air communication. (M-V3)

7.1.4.2 Internal Interface between On-Board FRMCS Gateway Function and On-Board FRMCS Radio Function

7.1.4.2.1 OB_{RAD}

7.1.4.2.1.1 Description

7.1.4.2.1.1.1 Interface OB_{RAD} is an implementation of the OB_{RAD} reference point. (I)

7.1.4.2.1.1.2 Interface OB_{RAD} enables communication between On-Board FRMCS Gateway Function and On-Board FRMCS Radio Function(s). (I)

7.1.4.2.1.1.3 OB_{RAD} enables the objectives stated in clause [7.1.3.1.2]. (I)

7.1.4.2.1.1.4 OB_{RAD} is used to control and manage radio modules within the On-Board FRMCS Radio Function(s). (I)

7.1.4.2.1.2 Interface Requirements

7.1.4.2.1.2.1 OB_{RAD} shall enable Control Plane (Session) communication between On-Board FRMCS Gateway Function and On-Board FRMCS Radio Function(s). (M-V3)

7.1.4.2.1.2.2 OB_{RAD} shall enable User Plane (Media) communication between On-Board FRMCS Gateway Function and On-Board FRMCS Radio Function(s). (M-V3)

7.1.4.2.1.2.3 The OB_{RAD} Control Plane shall enable relocation of established communication session(s) between On-Board FRMCS Radio Functions. (M-V3)

7.1.4.2.1.2.4 OB_{RAD} Control Plane shall enable the selection and usage of On-Board FRMCS Radio Function(s). (M-V3)

7.1.4.2.1.2.5 OB_{RAD} Control Plane will enable the selection and usage of the Radio Module(s) hosted by On-Board FRMCS Radio Function(s). (I)

7.1.4.2.1.2.6 OB_{RAD} supports in-service replacement of On-Board FRMCS Radio Function(s). (I)

7.1.4.2.1.2.7 OB_{RAD} is implemented using suitable, current and widely adopted industry specifications. (I)

Note: OB_{RAD} control plane is unrelated to OB_{APP} control plane.

Note: OB_{RAD} interface specification may use [TR 104 006] as initial input.

7.1.4.2.1.3 Physical Interface between On-Board FRMCS and the FRMCS Radio Function(s) or Radio Module(s) (OBRAD physical interface)

7.1.4.2.1.3.1 Requirements

- 7.1.4.2.1.3.1.1 The OBRAD physical interface shall support distributed deployments of FRMCS Radio Functions within a train realizing the On-Board FRMCS detachable configuration as defined in [TOBA-FRS]. (M-V3)
- 7.1.4.2.1.3.1.2 The OBRAD physical interface shall enable sufficient performance (bandwidth, latency) to carry both the user plane traffic and the required control plane traffic without degrading the QoS of the Communication Services, as it is defined in section [14] of this document. (M)
- 7.1.4.2.1.3.1.3 The OBRAD physical interface shall be realised using a suitable industry standard fulfilling the functional requirements stated under section [7.11] of [TOBA-FRS]. (M)
- 7.1.4.2.1.3.1.4 The OBRAD reference point shall be able to use an existing network (e.g., CCS Consist Network as defined in [SUBSET-147]) that can provide required security and performance stated in sections [14] and [15] of this document with respect to bandwidth integrity and confidentiality. (M)

7.1.5 On-Board FRMCS Gateway Function

7.1.5.1 Introduction

7.1.5.2 An On-Board FRMCS Gateway Function encompasses Functions that enable the specification of (I):

- On-Board FRMCS system requirements related to communication service(s) enabled by FRMCS ([7.1.5.4], [7.1.5.5],[7.1.5.6], [7.1.5.7], [7.1.5.8], [7.1.5.9], [7.1.5.10]).
- System requirements related to the operation and maintenance of the On-Board FRMCS ([7.1.5.11.2.2]).

7.1.5.3 Communication Gateway Function

7.1.5.3.1 A Communication Gateway Function encompasses Functions related to communication service(s) for the On-Board FRMCS (I).

7.1.5.3.2 A Communication Gateway Function contains the following Functions with reference to Figure 7-2: (I):

- API ([7.1.5.4])
- Orchestration ([7.1.5.8])
- FRMCS Service Client(s) ([7.1.5.9])
- Connectivity ([7.1.5.10])

7.1.5.4 API

7.1.5.4.1 The API shall include the following functions (M):

- a. Local Binding
- b. Session Management
- c. Notification Function

7.1.5.4.2 The API is specified by [FRMCS-FFFIS]. (I)

7.1.5.5 Local Binding

7.1.5.5.1 Introduction

7.1.5.5.1.1 The Local Binding function is a function of the API. (I)

7.1.5.5.1.2 The Local Binding Function enables authorized data flows and provides means to prevent unauthorized data flow(s). (I)

7.1.5.5.2 Requirements

7.1.5.5.2.1 For data flows associated to control plane of Application instances, Local Binding shall provide the following:

- a. Identification (M)
- b. Authentication (M)
- c. Integrity (M)
- d. Confidentiality (M)

7.1.5.5.2.2 Using Local Binding, the On-Board FRMCS shall have the capability to identify itself to Applications. (M)

7.1.5.5.2.3 Using Local Binding, the On-Board FRMCS shall have the capability to authenticate itself to Applications. (M)

7.1.5.5.2.4 For data flows associated to user plane of Application instances, Local Binding shall provide the following:

- a. Identification (M)
- b. Authentication (M)

Note: This is in order to be able to identify each data flow as an FRMCS Service Session and to prevent any unauthorized establishment of FRMCS Service Sessions.

7.1.5.5.2.5 For data flows associated to user plane of Application instances, Local Binding should enable the following:

- a. Integrity (O)
- b. Confidentiality (O)

Note: This is in order to provide means of data protection for Applications that have no such capability but have the need to achieve a certain level of protection.

7.1.5.5.2.6 With respect to Identification of Application instances Local Binding shall support an application registration process to the On-Board FRMCS [FRMCS-FFFIS]. (M)

7.1.5.5.2.7 The application registration process shall enable differentiation between the application coupling modes whether it is Loose or Tight as described in section [6.1.3.1]. (M)

7.1.5.5.2.8 With respect to Identification of Application instances, Local Binding shall comply with Service Session parameters. (M-V3)

Note: Technical details of Local Binding are further specified in the [FRMCS-FFFIS].

7.1.5.6 Session Management

7.1.5.6.1 Introduction

7.1.5.6.1.1 Session Management is a Function of the of API. (I)

7.1.5.6.1.2 Session Management enables, maintains or disables Communication Service Session(s). (I)

7.1.5.6.2 Requirements

7.1.5.6.2.1 Session Management shall enable communication service sessions between two single users. (M)

7.1.5.6.2.2 Session Management shall enable communication service sessions concurrently between any one single user and multiple users. (M)

7.1.5.6.2.3 Session Management shall enable unicast communication. (M)

7.1.5.6.2.4 Intentionally deleted.

7.1.5.6.2.5 Session Management shall include the capability to establish communication service sessions. (M)

7.1.5.6.2.6 Session Management shall include the capability to inform applications about the status of communication service sessions. (M)

7.1.5.6.2.7 Session Management shall include the capability to release communication service sessions. (M)

7.1.5.6.2.8 Session Management shall include the capability to uniquely identify communication service sessions. (M)

7.1.5.6.2.9 Session Management shall only activate communication service sessions for Application instances identified and authorized by Local Binding. (M)

7.1.5.7 Notification Function

7.1.5.7.1 Introduction

7.1.5.7.1.1 The Notification Function is a function of the API as specified in [FRMCS-FFFIS] chapter §9.11. (I)

7.1.5.7.1.2 The Notification Function manages notifications towards the On-board applications [FRMCS-FFFIS] (I).

7.1.5.7.2 Requirements

7.1.5.7.2.1 The Notification Function shall have the capability to provide information to the application using push and/or pull mechanism. (M)

7.1.5.7.2.2 The Push mechanism shall allow immediate notification on unforeseen situations (e.g., loss of signal, etc.). (M)

7.1.5.7.2.3 The Notification Function shall provide read-only information. (M)

7.1.5.7.3 Service capabilities

7.1.5.7.3.1 The Notification Function shall have the capability to send notification information elements (NOTIF-IE) to locally bound Applications using the notification services as specified in [FRMCS-FFFIS]. (M)

7.1.5.7.3.2 The Notification Function notification feature as specified in [FRMCS-FFFIS] shall be the only feature used to send NOTIF-IE in the direction from an On-Board FRMCS to a locally bound Application. (M)

7.1.5.7.3.3 Intentionally deleted.

7.1.5.7.3.4 The provisions for sending NOTIF-IE from an On-Board FRMCS by the Notification Function using the Notification Function notification feature shall be configurable as specified in [FRMCS-FFFIS]. (M)

Note: With configurable is meant e.g. the Parameter name 'Update period for the Notification Function'.

7.1.5.7.3.5 The Notification Function shall have the capability to produce the following categories of NOTIF-IE, as specified in clause [7.1.5.7.7] (M):

- a. Availability of FRMCS Transport Domain for FTD_AVL.
- b. Non-availability of FRMCS Transport Domain for FTD_NAVL_NOTIF.
- c. Availability of FRMCS Service Domain for FSD_AVL.
- d. Non-availability of FRMCS Service Domain for FSD_AVL.

Note: Network transition state changes corresponds with Category of Notification Function information: communication Status in [FRMCS-FFFIS].

7.1.5.7.3.6 The Notification Function shall have the capability to produce the following categories of NOTIF-IE:

- a. Train Location Information as defined in section 16.3.16 (M)
- b. FRMCS Time Service as defined in chapter 23 (M-Vx)
- c. Observed Indicative QoS (M-Vx)
- d. Subscribers of identified types (M-Vx)

7.1.5.7.3.6i For Tight Coupled Applications, the Notification Function:

- a. Shall be able to convey Train Location Information (M)
- b. Shall be able to convey FRMCS Time Service (M-Vx)
- c. Shall be able to convey Observed Indicative QoS (M-Vx)
- d. Shall be able to convey Subscribers of identified types (M-Vx)

7.1.5.7.3.6ii For Loose Coupled Applications, the Notification Function:

- a. Should be able to convey Train Location Information (O)

- b. Shall be able to convey FRMCS Time Service (M-Vx)
- c. Shall be able to convey Observed Indicative QoS (M-Vx)
- d. Shall be able to convey Subscribers of identified types (M-Vx)

Editor's note: The correspondence between NOTIF-IE in the above list and Category of Notification Function as specified in [FRMCS-FFIS] and [TOBA-FRS] is not yet completed.

7.1.5.7.3.7 Any NOTIF-IE message generated by the Notification Function and sent using Notification Function notification feature to an Application shall use the appropriate 'Value of Notification Function information' that corresponds with 'Category of Notification Function information' as specified in [FRMCS-FFIS]. (M)

7.1.5.7.3.8 The correspondence of Category of Notification Function information with Value of Notification Function information is listed in [FRMCS-FFIS]. (I)

7.1.5.7.4 Reliability

7.1.5.7.4.1 The reliability of the Notification Function is depending on the reliability of the data sources (e.g. Time Function or Location and Positioning Function) providing information elements as input to the Notification Function.(I)

7.1.5.7.5 Availability

7.1.5.7.5.1 The availability of the Notification Function is depending on availability of the data sources (e.g. Time Function or Location and Positioning Function) providing information elements as input to the Notification Function. (I)

7.1.5.7.6 Maintenance and troubleshooting of the Notification Function

7.1.5.7.6.1 This section concerns requirements for the purpose of providing mechanisms for trouble shooting and/or quality assurance of the Notification Function. (I)

7.1.5.7.6.2 For all sources participating in NOTIF-IE production, the Notification Function shall have the capability to monitor, maintain current status and detect changes in status of availability and non-availability. (M)

Note: The means to achieve this capability is considered out of scope of this chapter.

7.1.5.7.6.3 The Notification Function shall have the capability to produce log records for detected changes of availability status including timestamp and additional data if available. (M)

7.1.5.7.6.4 The Notification Function should have the capability to maintain a set of the latest obtained data from all sources. (O)

7.1.5.7.6.5 If the Notification Function is maintaining a set of the latest obtained data, then for all log records produced, the most recently introduced or updated record of this maintained data set shall be included as additional data associated with any log record produced. (M)

7.1.5.7.6.6 Based on a maintainable, configurable (on/off) mechanism and for all NOTIF-IE categories generated, the Notification Function should provide the capability to send the following information to the OM (O):

- Timestamp
- Train Location Information
- Application Category [FRMCS-FFFIS]
- Application On-Board identifier [FRMCS-FFFIS]
- Identifier of a session (if applicable) [FRMCS-FFFIS]
- NOTIF-IE
- Indication of successful/unsuccessful sending

7.1.5.7.7 Notifications on FRMCS availabilities

7.1.5.7.7.1 The Notification Function shall be able to notify the state transition from FTD_AVL to FTD_NAVL to an application (see section 7.1.5.12), i.e., through Notification Services defined in [FRMCS-FFFIS]. (M)

7.1.5.7.7.2 The notification in clause 7.1.5.7.7.1 is referred, as FTD_NAVL_NOTIF. (I)

7.1.5.7.7.3 The Notification Function shall be able to notify the state transition from FTD_NAVL to FTD_AVL to an application (see section 7.1.5.12), i.e., through Notification Services defined in [FRMCS-FFFIS]. (M)

7.1.5.7.7.4 The notifications in clause 7.1.5.7.7.3 is referred, as FTD_AVL_NOTIF. (I)

7.1.5.7.7.5 The FTD_AVL_NOTIF and FTD_NAVL_NOTIF messages shall include the reason of notification (e.g., “Network Transition”). (M)

7.1.5.7.7.6 If the reason of FTD_AVL_NOTIF is “Network Transition”, the notification message shall include the Target FRMCS Domain. (M)

7.1.5.7.7.7 The Notification Function shall be able to notify the state transition from FSD_AVL to FSD_NAVL to an application (see section 7.1.5.12), i.e., though Notification Services defined in [FRMCS-FFFIS]. (M)

7.1.5.7.7.8 The notifications in clause 7.1.5.7.7.7 is referred as FSD_NAVL_NOTIF. (I)

7.1.5.7.7.9 The Notification Function shall be able to notify the state transition from FSD_NAVL to FSD_AVL to an application (see section 7.1.5.12), i.e., though Notification Services defined in [FRMCS-FFFIS]. (M)

7.1.5.7.7.10 The notifications in clause 7.1.5.7.7.9 is referred as FSD_AVL_NOTIF. (I)

7.1.5.7.7.11 The FSD_AVL_NOTIF and FSD_NAVL_NOTIF messages shall include the reason of notification (e.g., “Network Transition”). (M)

7.1.5.8 Orchestration

7.1.5.8.1 Introduction

7.1.5.8.1.1 The Orchestration Function is a function of the Communication Gateway. (I)

7.1.5.8.1.2 The Orchestration Function manages the routing of user plane data flow inside the communication gateway depending on the coupling mode (Loose or Tight [TOBA-FRS] and [FRMCS-FFFIS] chapters 5.6 and 5.7). (I)

7.1.5.8.2 Requirements

- 7.1.5.8.2.1 The Orchestration function shall have the capability to use data from Local Binding about the coupling mode (Loose or Tight [TOBA-FRS]). (M)
- 7.1.5.8.2.2 The orchestration function shall have the capability to route user plane dataflows associated to locally bound Loose-Mode application instances to the appropriate FRMCS Service Client. (M)
- 7.1.5.8.2.3 The Orchestration Function shall manage the routing of user plane data flow inside the communication gateway using information derived from (M):
- Local Binding
 - Session Management
- 7.1.5.8.2.4 The information used for the routing of user plane data shall encompass (M):
- Coupling Mode
 - Application Identifier
 - Session Identifier
 - Type of communication
- 7.1.5.8.2.5 The routing of user plane data flow shall be implemented in a way that enables support for and QoS treatment of the applications according to Annex A, Table A.1. (M)
- 7.1.5.9 FRMCS Service Client(s)
- 7.1.5.9.1 Introduction
- 7.1.5.9.1.1 MCX for FRMCS framework is specified in [TS 103 765-2]. (I)
- 7.1.5.9.2 Requirements
- 7.1.5.9.2.1 The Communication Gateway shall host one or more FRMCS Service Clients to enable Loose Coupled communication mode. (M)
- 7.1.5.9.2.2 Intentionally deleted.
- 7.1.5.9.2.3 FRMCS Service Clients manage communication service session(s) with the MCX server. (I)
- 7.1.5.10 Connectivity
- 7.1.5.10.1 Introduction
- 7.1.5.10.1.1 Connectivity is a function of the Communication Gateway. (I)
- 7.1.5.10.1.2 Connectivity concerns data flows in the OB_{APP} user plane. (I)
- 7.1.5.10.2 Requirements

- 7.1.5.10.2.1 Connectivity shall enable single path communication. (M)
- 7.1.5.10.2.2 Connectivity shall contain the following function: FRMCS Multipath Function. (M-V3)
- 7.1.5.10.2.3 Connectivity shall provide dataflow transport resources. (M)
- 7.1.5.10.2.4 Connectivity shall manage and control On-Board FRMCS Radio Function(s) using OB_{RAD} if implemented. (M-V3)
- 7.1.5.10.3 The On-Board FRMCS Multipath
 - 7.1.5.10.3.1 The On-Board FRMCS Multipath is a function of Communication Gateway Function. (I)
 - 7.1.5.10.3.2 The On-Board FRMCS Multipath is a function that manages and controls concurrent user plane data flow distribution over OB_{RAD}. (I)
Note: See definition of concurrency and chapter [12] for more details.
 - 7.1.5.10.3.3 The On-Board FRMCS Multipath Function is specified in chapter [12] of this document. (I)
 - 7.1.5.10.3.4 The On-Board FRMCS Multipath Function will be configurable by an MPF policy received from an MPF on the infrastructure side ([12.3.27] point 3). (I)
 - 7.1.5.10.3.5 The On-Board FRMCS Multipath Function will, according to the MPF policy, be able to transfer the data flow of a given application through a selected data path ([12.3.19], [12.3.20]). (I)

7.1.5.11 OM

7.1.5.11.1 Introduction

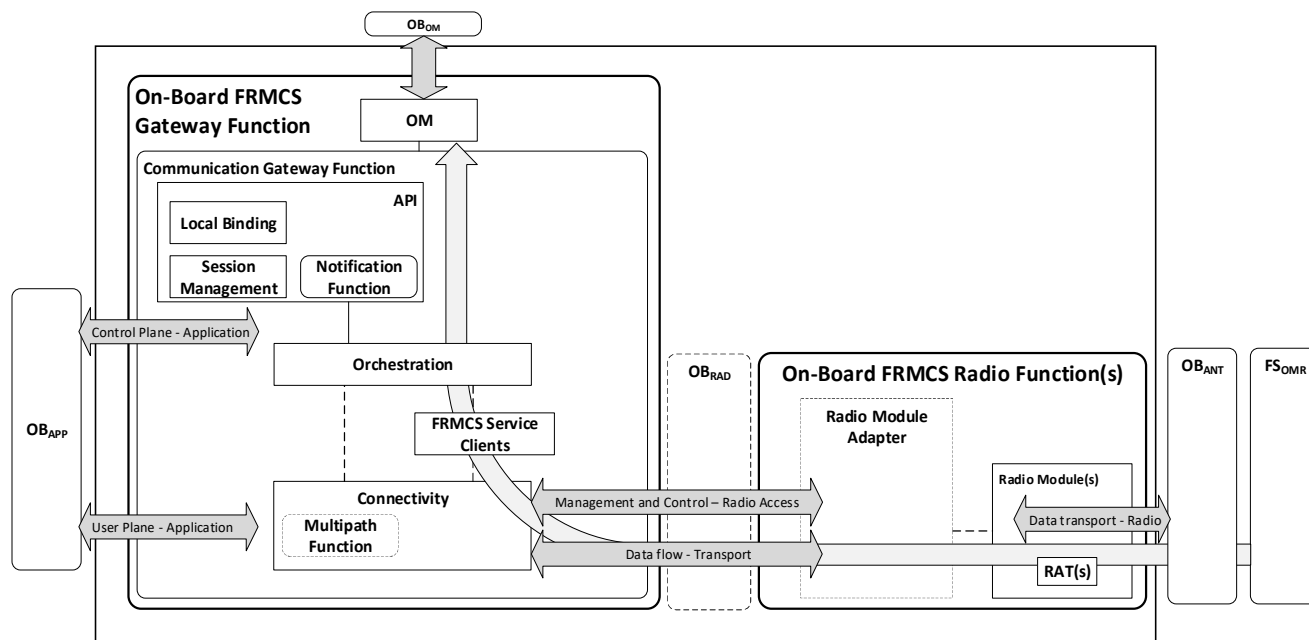


Figure 7-3: On-Board FRMCS architecture with OM

7.1.5.11.1.1 OM is a function of Gateway Function. (I)

7.1.5.11.1.2 OM is related to operation and maintenance of the On-Board FRMCS. (I)

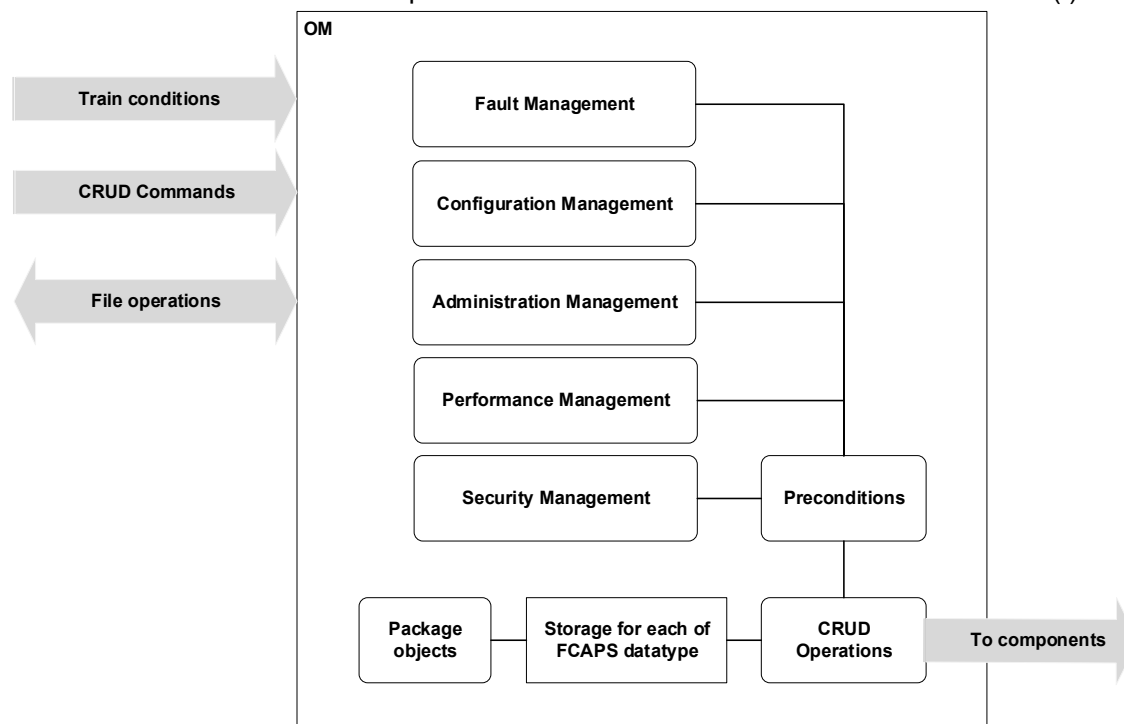


Figure 7-4: System view of OM function

7.1.5.11.2 Requirements

7.1.5.11.2.1 OM Common Requirements

- 7.1.5.11.2.1.1 OM shall enable Operation and Maintenance of On-Board FRMCS. (M)
- 7.1.5.11.2.1.2 The OM shall enable access to Operation and Maintenance functionalities via OB_{OM} interface (local OM access). (M)
- 7.1.5.11.2.1.3 Local OM access shall be available with and without available OTA connectivity. (M)
- 7.1.5.11.2.1.4 The OM shall enable access to Operation and Maintenance functionalities via FS_{OMR} interface (OTA OM access). (M-V3)
- 7.1.5.11.2.1.5 OTA OM access using an interface over FS_{OMR} reference point is expected to use OB_{ANT} and OB_{RAD} interfaces to obtain connectivity to the OM hosted by an On-Board FRMCS. (I)
- 7.1.5.11.2.1.6 Multiple concurrent OM Sessions should be possible. (O)

7.1.5.11.2.2 OM Capabilities

- 7.1.5.11.2.2.1 The OM shall collect internal log records and data. (M)
- 7.1.5.11.2.2.1i The OM shall be able to receive and transfer information to external applications (local and remote) by means of a fully documented API that makes use of a standard protocol for communication (e.g, http, etc..). (M)
Note: “fully documented” in this context implies that documentation is published by the organisation that develops the API, such that it is possible for third parties to develop applications based on the same API without need to acquire additional documentation or information.
- 7.1.5.11.2.2.2 The OM API implements specified OM functionalities. (I)
- 7.1.5.11.2.2.3 The OM shall have the capability to consider the train operational status based on external information provided via OB_{OM} interface. (M-Vx)
- 7.1.5.11.2.2.3i The OM shall have the capability to consider the train operational status based on external information using FS_{OMR} interface. (M-Vx)
- 7.1.5.11.2.2.4 The OM shall have the capability of identifying train operational status based on detection mechanism(s). (M-V3)
- 7.1.5.11.2.2.5 The On-Board FRMCS shall have the capability to determine the possibility of entering (or not) in a specific On-Board FRMCS maintenance status based on OBF OM-SCP. (M-V3)
- 7.1.5.11.2.2.6 The maintenance statuses are as per the table 7-1 below. (I)

On-Board FRMCS Maintenance Status	Description
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Start of Operations	Train is at the preparation phase of its operation, systems are booting up, initial information exchange are taking place
Normal Operation	Train is in normal operation, OB FRMCS is connected and some applications (Voice, ETCS, ...) may be currently used
OBF Maintenance	OB FRMCS is under maintenance -undergoing a configuration, update or upgrade
GW SW Update	OB FRMCS GW is currently undergoing a software update
GW SW Upgrade	OB FRMCS GW is currently undergoing a software upgrade
Individual OBF RF SW Update	One On-Board FRMCS Radio Function is currently being updated -status need to be attached to a specific ID of the OBF RF-
Individual OBF RF SW Upgrade	One On-Board FRMCS Radio Function is currently being upgraded -status need to be attached to a specific ID of the OBF RF-
Individual OBF RM SW Update	One On-Board FRMCS Radio Module is currently being updated -status need to be attached to a specific ID of the OBF RF and OBF RM-
Individual OBF RM SW Upgrade	One On-Board FRMCS Radio Module is currently being upgraded -status need to be attached to a specific ID of the OBF RF and OBF RM-
Connectivity-affecting parameter configuration	O&M user has initiated the change of at least one parameter having an impact on connectivity (e.g., eSIM, credentials, ...)
Non-Connectivity-Affecting parameter configuration	O&M user has initiated the change of at least one parameter having no impact on connectivity (e.g., application profiles, ...)
Log retrieval status (background)	O&M user has initiated a query for a specific log
Audit Log retrieval status (background)	Audit log retrieval status (background): O&M user has initiated a query for a specific audit log

Table 7-1 On-Board FRMCS Maintenance Status (based on OBF OM-SCP)

7.1.5.11.2.2.7 The OM shall provide the capability of configuring combinations of conditions on the OBF OM-SCP allowing the activation of specific On-Board FRMCS maintenance statuses. (M-V3)

7.1.5.11.2.2.8 The On-Board FRMCS shall have the capability to change the On-Board FRMCS Maintenance Status only if the configured combination of conditions for the target status is verified. (M-V3)

7.1.5.11.2.3 OM Fault Management

7.1.5.11.2.3.1 The On-Board FRMCS system shall detect and classify defects within the On-Board FRMCS boundary that affect its normal operation. (M)

7.1.5.11.2.3.2 The On-Board FRMCS system shall create and send an alert and error message immediately about defects that could prevent its normal operation over the O&M interface OB_{OM}. (M)

7.1.5.11.2.3.2i The On-Board FRMCS system shall create and send an alert and error message immediately about defects that could prevent its normal operation over the O&M interface FS_{OMR}. (M-V3)

7.1.5.11.2.3.3 The monitored parameter set via the O&M function should support detecting anomalies before they develop into actual defects. (O)

7.1.5.11.2.3.4 In case of a malfunction and/or resource exhaustion, the On-Board FRMCS system should deallocate resources ([7.1.5.11.2.6.1]) associated to non-critical service sessions in case this allows the allocation of necessary resources for sustained operation of higher-priority service sessions ([FRMCS FRS] chapter 8.2.8). (O)

7.1.5.11.2.3.5 The On-Board FRMCS shall detect malfunctioning Hardware and Software components ([7.1.5.11.2.6.2]) and disable them such that they are no longer operational, without impacting critical functionalities. (M-Vx)

7.1.5.11.2.3.6 The On-Board FRMCS shall perform corrective actions (e.g., rebooting, reconfiguration, health check, etc) on individual malfunctioning components in such a way that it does not impact critical functionality. (M-Vx)

7.1.5.11.2.3.7 Only if the corrective actions taken according to [7.1.5.11.2.3.6] can result in full recovery of the malfunctioning component, the component may be used again. (I)

7.1.5.11.2.4 OM Configuration Management

7.1.5.11.2.4.1 The OM shall enable configuration management of the On-Board FRMCS. (M)

7.1.5.11.2.4.2 The OM shall enable authenticated and authorized O&M Users to perform any one of the Create, Read, Update, Delete (CRUD) operations on the FRIOP information elements, see chapter [19.2]. (M)

7.1.5.11.2.4.3 The OM shall enable authenticated and authorized O&M Users to perform any one of the CRUD operations on the non-FRIOP information elements, see chapter [19.2]. (M)

7.1.5.11.2.4.4 The OM shall enable On-Board FRMCS Package management capabilities for On-Board FRMCS components. (M)

Note: As per definition of On-Board FRMCS Package, this can refer to a single On-Board FRMCS component, to multiple components, or to the whole On-Board FRMCS.

7.1.5.11.2.4.5 The OM shall enable O&M Users to perform any one of the CRUD operations on On-Board FRMCS Package. (M)

7.1.5.11.2.4.6 On-Board FRMCS Package management capabilities should encompass storage of On-Board FRMCS Packages in a storage facility located in the On-Board FRMCS. (O)

7.1.5.11.2.4.7 In case a storage facility for On-Board FRMCS On-Board FRMCS Packages is available, Package management capabilities shall enable any of the CRUD operations on stored On-Board FRMCS Packages. (M)

7.1.5.11.2.4.8 On-Board FRMCS Package management shall encompass package versioning with compatibility check for On-Board FRMCS Packages. (M)

7.1.5.11.2.4.9 On-Board FRMCS Package management capabilities shall have the capability to configure sources for On-Board FRMCS Packages. (M)

7.1.5.11.2.4.10 Each source for On-Board FRMCS Packages is expected to be secured. (I)

7.1.5.11.2.4.11 On-Board FRMCS Package management capabilities shall have the capability to verify the authenticity of the On-Board FRMCS Packages. (M)

7.1.5.11.2.4.12 The OM shall provide the capability of maintaining a stored current list of system components of the On-Board FRMCS including their unique IDs. (M)

7.1.5.11.2.4.12i The stored current list of system components as specified in [7.1.5.11.2.4.12] shall as a minimum consist of and by means of reference be associated with the architecture elements described and specified within chapter [7.1] of this specification: (M)

Note: With by means of reference is in this context to be understood as the capability to relate entities in a product with architectural elements in related specification(s).

7.1.5.11.2.4.12ii The content and details for the stored current list of system components shall comply to [7.1.5.11.2.4.12] (M-V3)

7.1.5.11.2.4.13 The OM shall provide the capability of retrieving the list of all the identifiable elements of the On-Board FRMCS including their unique IDs. (M)

7.1.5.11.2.4.14 The OM should provide the capability of maintaining a protected stored set of all the current parameters for the FRMCS service clients hosted by the On-Board FRMCS. (O)

7.1.5.11.2.4.15 The OM should provide the capability only to authenticated and authorized O&M Users to perform any one of the CRUD operations on information elements stored in the current parameter storage. (O)

7.1.5.11.2.5 OM Administration

7.1.5.11.2.5.1 The OM shall enable protected access for O&M Users granted with administrative rights to perform any one of the CRUD operations on OM user accounts. (M)

7.1.5.11.2.5.2 The OM shall enable protected access for O&M Users granted with administrative rights to perform any one of the CRUD operations on OM user groups. (M)

7.1.5.11.2.5.3 The OM should enable protected administrative access to configuration of assignment of OM users to OM user groups. (O)

7.1.5.11.2.5.4 The OM shall enable protected administrative access to configuration of OM user access rights. (M)

7.1.5.11.2.5.5 The OM shall grant access to O&M Users exclusively based on configured access rights. (M)

7.1.5.11.2.5.6 The OM shall provide the capability to distinguish access rights based on the type of access (local OM access or OTA OM access). (M)

7.1.5.11.2.5.7 O&M Users shall be able to perform operations and maintenance activities on On-Board FRMCS components with their associated configuration parameters. (M)

Editor's Note: A matrix of recommended access for O&M User rights related to FRIOP, non-FRIOP and Cybersecurity parameters could be provided in a later stage but not in scope of FRMCS v2.

7.1.5.11.2.5.8 The OM shall provide the capability to store a current list of all the On-Board FRMCS entities authorized to use Local Binding. (M)

7.1.5.11.2.5.8i The current list of all the On-Board FRMCS entities as specified by [7.1.5.11.2.5.8] shall be maintainable. (M)

7.1.5.11.2.5.9 The OM shall enable protected administrative access to perform any one of the CRUD operations on Local Binding credentials for FRMCS entities. (M)

7.1.5.11.2.5.10 OM should enable protected administrative access to configuration of Local Binding credentials for FRMCS entities. (O)

7.1.5.11.2.5.11 The OM shall provide the capability of maintaining a protected stored set of all the current service credentials for the FRMCS service clients hosted by the On-Board FRMCS. (M)

7.1.5.11.2.5.12 The OM shall enable protected administrative access to perform any one of the CRUD operations on records of service credentials in the current set. (M)

7.1.5.11.2.5.13 The OM shall provide the capability of maintaining a protected stored repository for transport stratum-related identifiers and parameters for:

- Radio Access (e.g., eSIM Card) (M)
- IMS/SIP Core access (M)

7.1.5.11.2.5.14 The OM shall enable protected administrative access to perform any one of the CRUD operations on records into the current repository of transport stratum-related identifiers and parameters. (M)

7.1.5.11.2.6 OM Performance Monitoring

7.1.5.11.2.6.1 The On-Board FRMCS shall monitor performance indicators related to On-Board FRMCS HW resources: (M)

- CPU load (including monitoring load)
- Memory usage
- Storage usage
- Networking load
- Networking usage type (e.g., DPI)

7.1.5.11.2.6.2 The On-Board FRMCS shall monitor performance indicators related to On-Board FRMCS components: (M)

- Gateway Function
- On-Board FRMCS Radio Function(s)
- Radio Module(s)
- SW Components
- HW Components

7.1.5.11.2.6.3 The On-Board FRMCS shall monitor performance indicators related to On-Board FRMCS interfaces: (M)

- OB_{APP}
- OB_{OM}

7.1.5.11.2.6.4 The On-Board FRMCS should monitor performance indicators related to On-Board FRMCS interfaces: (O)

- OB_{RAD}
- OB_{ANT}

7.1.5.11.2.6.5 The On-Board FRMCS should have the capability to generate performance event log records based on configurable thresholds for the monitored performance indicators. (O)

7.1.5.11.2.6.6 The On-Board FRMCS shall enable authenticated and authorised O&M users to retrieve the performance event log records relative to the monitored indicators. (M)

7.1.5.11.2.7 OM Supervision and Audit

7.1.5.11.2.7.1 The On-Board FRMCS shall include parameters that as a set, uniquely can identify a source for all generated log records. (M)

7.1.5.11.2.7.2 The On-Board FRMCS shall include time stamping in all generated log records according to chapter [23.4] of this document. (M)

7.1.5.11.2.7.3 The On-Board FRMCS shall have the capability to include location information in all generated log records according to FRMCS Location and Positioning requirements according to chapter [16.3] of this document. (M)

7.1.5.11.2.7.3i The capability as specified by [7.1.5.11.2.7.3] shall be applied whenever there is a FRMCS Service session established for any Application of Type I or Type III as defined by [10.1.3.3.4]. (M)

7.1.5.11.2.7.3ii The capability as specified by [7.1.5.11.2.7.3] should be applied whenever there is a FRMCS Service session established for any Application of Type II or Type IV as defined by [10.1.3.3.4]. (O)

7.1.5.11.2.7.4 The On-Board FRMCS shall provide the capability of protecting selected log records from alterations. (M)

7.1.5.11.2.7.5 The On-Board FRMCS shall provide the capability of configuring which log records that shall be protected from alteration. (M)

7.1.5.12 States within On-Board FRMCS

7.1.5.12.1 The On-Board FRMCS shall be able to identify in which of the following state(s) it is operating (M):

- a) FRMCS Transport Domain Available (FTD_AVL)
- b) FRMCS Transport Domain Not-Available (FTD_NAVL)
- c) FRMCS Service Domain Available (FSD_AVL)
- d) FRMCS Service Domain Not-Available (FSD_NAVL)

Note: There is one FSD-AVL / FSD_NAVL per FRMCS Service User.

7.1.5.12.2 The On-Board FRMCS shall detect the transitions between FTD_NAVL and FTD_AVL. (M)

7.1.5.12.3 The On-Board FRMCS shall detect the transitions between FSD_NAVL and FSD_AVL, per FRMCS Service User. (M)

7.1.5.12.4 The detailed definition of states in clause [7.1.5.12.1] shall comply with [TS 103 765-3]. (M)

7.1.6 On-Board FRMCS Radio Function

7.1.6.1 Introduction

7.1.6.1.1 **Intentionally deleted.**

7.1.6.1.2 The On-Board FRMCS Radio Function includes the following components (I):

- Radio Module Adapter
- Radio Module(s)

7.1.6.1.2i The On-Board FRMCS Radio Function shall enable access for the Communication Gateway Function to the FRMCS Transport Stratum. (M)

7.1.6.1.2ii The On-Board FRMCS Radio Function enables transmission of control and user plane related data. (I)

7.1.6.1.2iii The On-Board FRMCS Radio Function boundaries are given by reference points OB_{RAD} and OB_{ANT}. (I)

7.1.6.1.3 **Intentionally deleted.**

7.1.6.1.4 **Intentionally deleted.**

7.1.6.1.5 **Intentionally deleted.**

7.1.6.2 Radio Module Adapter

7.1.6.2.1 The Radio Module Adapter is a function of the On-Board FRMCS Radio Function. (I)

7.1.6.2.2 The Radio Module Adapter enables On-Board FRMCS integrated architectures. (I)

7.1.6.2.3 The Radio Module Adapter enables On-Board FRMCS distributed architectures. (I)

7.1.6.2.4 The Radio Module Adapter enables adaptation of generalized and standardized control and management commands to manufacturer specific radio module commands. (I)

7.1.6.2.5 The Radio Module Adapter is meant to support interchangeability as defined in the definitions section. (I)

7.1.6.2.6 The Radio Module Adapter encompasses hardware (e.g., connectors). (I)

7.1.6.2.7 The Radio Module Adapter can encompass software. (I)

7.1.6.2.8 The Radio Module Adapter shall interface to OB_{RAD}. (M-V3)

7.1.6.2.9 The Radio Module Adapter can manage connectivity to one or more radio modules. (I)

7.1.6.2.10 The Radio Module Adapter interfaces with Radio Modules using suitable industrial interfaces (I)

7.1.6.3 Radio Module(s)

7.1.6.3.1 The On-Board FRMCS supports several types of Radio Modules. (I)

7.1.6.3.2 Radio Module types are for example (non-exhaustive list): (I)

- a. On-Board FRMCS Radio Module
- b. MNO Radio Module
- c. WLAN (e.g., Wi-Fi) Radio Module

7.1.6.3.3 **Intentionally deleted.**

7.1.6.3.4 A Radio Module has one or more external antenna interface(s), for Terrestrial Networks (TN) or Non-Terrestrial Networks (NTN) for radio access that can be either bi-directional or unidirectional. (I)

7.1.6.3.5 The On-Board FRMCS Radio Module shall comply to [TS 103 765-3]. (M)

7.1.6.3.6 The On-Board FRMCS Radio Module shall comply to requirements defined in chapter [8] of this document. (M)

7.1.6.3.7 **Intentionally deleted.**

7.1.6.3.8 For an On-Board FRMCS Radio Module, it shall be possible to disable or enable each band individually. (M-Vx)

7.1.6.3.9 For ITU region 1 and within the European Economic Area (EEA), the On-Board FRMCS Radio Module shall be capable of operating in the radio environment as defined identified per [ECC Decision (20)02]. (M)

7.1.6.3.10 The On-Board FRMCS Radio Module shall allow the implementation of National Roaming for selected railway lines. (M-V3)

7.1.6.3.11 For the purpose of National Roaming, the On-Board FRMCS Radio Module shall enable non-simultaneous usage of either RMR or MNO frequencies. (M-V3)

7.1.6.3.12 The On-Board FRMCS Radio Module shall allow the implementation of Multi-Operator Core Network (MOCN) for selected railway lines. (M-V3)

7.1.6.3.13 The MNO Radio Module shall support the frequency bands as listed in Table 8-1 of section [8] of this document. (M-V3)

7.1.6.3.13i The MNO Radio Module shall support the frequency bands as listed in Table 8-2 of section [8] of this document. (M-V3)

7.1.6.3.13ii The MNO Radio Module shall support the frequency bands as listed in Table 8-3 of section [8] of this document. (M-V3)

7.1.6.3.13iii The MNO Radio Module shall support the frequency bands as listed in Table 8-4 of section [8] of this document. (M-V3)

7.1.6.3.13iii The MNO Radio Module shall support the frequency bands as listed in Table 8-5 of section [8] of this document. (M-V3)

7.1.6.3.14 The MNO Radio Module shall comply to requirements defined in section [8.5] of this document. (M)

7.1.6.3.15 For an MNO Radio Module, it shall be possible to disable or enable each band individually as per national configuration. (M-Vx)

7.1.6.3.16 On an OnBoard FRMCS supporting IFDT, the On-Board FRMCS Radio Module shall support carrying communications of FRMCS Service Users which are served by different FRMCS Service Domains (see chapter [10] for requirements on IFDT and interconnected FRMCS Service Domains). (M)

7.2 Antenna Function

7.2.1 Introduction

7.2.1.1 The Antenna Function is outside the boundary of the On-Board FRMCS as depicted in Figure 7-1. (I)

7.2.1.2 The Antenna Function is accessible via the OB_{ANT} interface specified in clause 7.1.4.1.2. (I)

7.2.1.3 The Antenna Function can contain RF combining and switching functions. (I)

7.2.2 Generic requirements

7.2.2.1 A minimum of two antenna elements shall be used for On-Board FRMCS Radio Functions (M):

- Antenna element 1 used for TX/RX
- Antenna element 2 used for RX or TX/RX

7.2.2.2 The antenna elements are implemented as individual antennas or several antenna elements in one housing (MIMO antennas). (I)

7.2.2.3 The physical setup of the antennas on the roof supports both correct functioning of each antenna and sufficient isolation between active radio systems onboard. (I)

Note: For the onboard system, to ensure the required performance of the critical RMR services, a sufficient isolation between radio systems operating simultaneously is necessary to prevent onboard interference effects (e.g., due to blocking and out-of-band emissions). A proper isolation shall be ensured between GSM-R and FRMCS in the RMR bands as well as between GSM-R and FRMCS and, if used, MNO bands. Isolation requirements for the simultaneous operation of RMR and MNO services are presented in Chapter 8.5. Measures like additional filters or disabling specific MNO bands can help to achieve the necessary isolation between the radio systems.